Output Form (OF)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: *DRBENCHMARK* | ***Context:*** *French Hospital Clinical NLP — Metropolitan France (DrBenchmark)*

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

> Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Output Form was flagged MODERATE priority. The benchmark uses well-defined, reproducible scoring protocols: SeqEval with IOB2 format and first-token prediction for sequence labeling [Q52, Q53], Spearman correlation and EDRM for STS [Q54], standard classification metrics with majority-class baselines [Q56], 4-run averaging with Student's t-test [Q51], and full hyperparameter documentation [Q49, Q99, Q100]. These are well-matched to typical NER/POS/MLC pipeline outputs (E3C-D5, E3C-D6 confirm IOB2 alignment). However, the deployment requires three output forms the benchmark does not measure: (1) confidence-scored multi-label outputs, (2) uncertainty flagging for high-uncertainty entities, (3) ranked candidate lists for multi-morbid case adjudication. None of these is directly evaluable within the benchmark's scoring framework — DEFT-2021 uses binary 0/1 one-hot label vectors (DEFT2021-D11), DiaMed uses single-label per case (DIAMED-D10), and STS averages annotator scores discarding disagreement signal (DEFT2020-D16, CLISTER-D8). Several datasets exhibit severe class imbalance that masks minority-class performance under standard metrics: PxCorpus medical_prescription ~89% vs replace <1% and negate ~2% (PXCORPUS-D14, PXCORPUS-D15); CAS cls neutral ~74% (CAS-D21); ESSAI cls neutral ~77% (ESSAI-D13). For prescription correction intents and clinical negation/speculation distinctions, standard accuracy will hide deployment-relevant failure modes. The benchmark's finding that 'all the models based on CamemBERT face difficulties in corpora with limited amount of data, such as MantraGSC Patents, where they fail to generate labels other than O' [Q73] is itself a manifestation of how standard metrics interact with imbalance. The output form is structurally aligned for sequence-labeling and standard classification but does not measure the calibration/uncertainty/ranking behavior central to the deployment.

ID	Evidence
Q49	“For further guidance, we have integrated all the training details, including hyperparameters, in Appendix A. This information will help users to reproduce and customize the experiments conducted with DrBenchmark.” (p.5)
Q51	“All the models are fine-tuned regarding a strict protocol using the same hyperparameters for each downstream task. The reported results are obtained by averaging the scores from four separate runs, thus ensuring robustness and reliability. We also report statistical significance computed using Student's t-test.” (p.5)
Q52	“To ensure a fair and consistent comparison among systems for sequence-to-sequence tasks such as POS tagging and NER, we chose the SeqEval (Nakayama, 2018) metric in conjunction with the IOB2 format and the training of all the models to predict only the label on the first token of each word as mentioned by Touchent et al. (2023).” (p.6)
Q53	“It provides a tokenizer-agnostic evaluation and mitigates any correlation between models' performances and the tokenization process.” (p.6)
Q54	“For STS tasks, the models' performance was assessed using two metrics: (1) the Spearman correlation, and (2) the mean relative solution distance accuracy (EDRM), as defined by the original authors of the DEFT-2020 dataset (Cardon et al., 2020).” (p.6)
Q56	“The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions.” (p.6)
Q73	“However, all the models based on CamemBERT face difficulties in corpora with limited amount of data, such as MantraGSC Patents, where they fail to generate labels other than 'O'.” (p.7)
Q99	“For the experiments, we utilize the following hyperparameters that yield optimal performance from the models.” (p.13)
Q100	“To mitigate overfitting, we locally save the best model based on its validation metric.” (p.13)
E3C-D5	"Cet aspect évoquait une tumeur solide du péritoine." tokens tagged [0,0,0,0,1,2,0,0,0] — Multi-token pathology span ("tumeur solide") tagged in IOB2 — confirms NER output form alignment
E3C-D6	"As she suffered from acute respiratory distress syndrome and required mechanical ventilation..." tags [0,0,0,0,1,2,2,2,...] — Multi-token disease span annotation in B/I scheme — confirms annotation consistency

CLISTER-D8	Le reste de l'examen échographique ne trouvait aucune autre anomalie." vs. "Le reste de l'examen somatique était sans anomalie. — Intermediate similarity; different exam modality with shared negative finding structure
DIAMED-D10	Le traitement était constitué d'énoxaparine d'antiagrégant plaquettaire, de diurétique de l'anse, de dérivés morphiniques et d'oxygène — Multi-system case forced into single digestive label — masks multi-morbidity
DEFT2021-D11	[1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 0.0, 1.0, 0.0, 0.0, 0.0, 1.0, 0.0, 1.0, 1.0, 0.0, 1.0] — Strictly binary 0/1 label vector — no confidence or soft-label dimension present
ESSAI-D13	Chaque cycle dure 28 jours — Trivially neutral sentence — inflates neutral class count, contributing to severe label imbalance
PXCORPUS-D14	c'est 2 pourcent le collyre — Surface reads like prescription; labeled negate — ambiguity in minority class
PXCORPUS-D15	negate=11, replace=3 in 500 sampled — Severe class imbalance; majority-class baseline masks minority-class performance
DEFT2020-D16	Après inspection, elles ont toutes été exclues. — Annotator disagreement averaged away in scoring; calibration signal lost
CAS-D21	buffer: negation_speculation:6, negation:100, neutral:368, speculation:26 — Heavy class imbalance; neutral dominates at ~74%

Strengths

- Reproducible unified protocol: 4-run averaging, Student's t-test significance, hyperparameters in appendix [Q51, Q99, Q100]
- Tokenizer-agnostic SeqEval IOB2 evaluation aligns with standard NER pipeline output formats [Q52, Q53, E3C-D5, E3C-D6]
- MIT-licensed code release supports deployment-side audit and reproducibility [Q95]
- Explicit majority-class baselines provide a floor for classification performance interpretation [Q56]

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Q56	“The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions.” (p.6)
Q95	“All code for DrBenchmark is released under the MIT License.” (p.9)
Q99	“For the experiments, we utilize the following hyperparameters that yield optimal performance from the models.” (p.13)
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E3C-D5	"Cet aspect évoquait une tumeur solide du péritoine." tokens tagged [0,0,0,0,1,2,0,0,0] — Multi-token pathology span ("tumeur solide") tagged in IOB2 — confirms NER output form alignment
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Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Output modality is text labels and sequence tags, matching deployment NLP outputs. However, the deployment additionally requires confidence scores, uncertainty flags, and ranked candidates — none of which are directly evaluated. The deployer's specified multi-label-with-confidence output behavior is not measurable with the current scoring framework [Q56, DEFT2021-D11].

OF-2: Not applicable — deployment is text-based with no speech output requirement.

OF-3: Not directly applicable — primary users (clinicians, nursing staff) have high domain expertise [elicitation Q2]; literacy is not a concern.

OF-4: External validity is harmed by absence of confidence calibration, uncertainty flagging, and ranked-candidate scoring (DEFT2021-D11, DIAMED-D10, DEFT2020-D16). Severe class imbalance in CAS, ESSAI, and PxCorpus (CAS-D21, ESSAI-D13, PXCORPUS-D15) means standard accuracy can mask poor minority-class performance on clinically critical correction intents and rare ICD-10 chapters. The benchmark's own observation that CamemBERT models produce only 'O' labels in low-resource scenarios [Q73] illustrates this mode of evaluation failure.

ID	Evidence
Q2	“We evaluate 8 state-of-the-art pre-trained masked language models (MLMs) on general and biomedical-specific data, as well as English specific MLMs to assess their cross-lingual capabilities.” (p.1)
Q56	“The results of the 8 models are reported in Table 6 and compared to a baseline obtained by considering the majority class for all predictions.” (p.6)
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Information Gaps

- Per-class F1 (not just macro/aggregate) and calibration curves are not reported in the benchmark for the deployment-critical minority classes

Requires Expert Verification

- What calibration thresholds (ECE, Brier score, AUROC per label) are operationally adequate for the deployer's clinician-adjudication workflow

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Remediation

Gap 1: Standard F1/Spearman/SeqEval scoring with majority-class baselines cannot measure calibration, uncertainty flagging, or ranked-candidate behavior; severe class imbalance in CAS/ESSAI/PxCorpus masks minority-class performance

Recommendation: Augment the benchmark's reporting with per-class F1, calibration plots, and ECE for all classification tasks. For PxCorpus minority classes (negate, replace) and CAS/ESSAI non-neutral labels, report F1 explicitly stratified by class to surface clinically critical failure modes in prescription correction and negation detection.